Second Volume.

First Book. Groups.

First Section. General Group Theory.

§ 1. Definition of Groups.

In the first volume, in connection with permutations, we became acquainted with the notion of a group and made important algebraic applications of it. Our next task must now be to formulate this concept, which is so exceedingly important in all more recent mathematics, in a more general way, and to become acquainted with the laws which govern it. We place the following definition at the beginning.

A system P of things, or elements, of any kind becomes a group if the following assumptions are satisfied:

1. A rule is given by which, from a first and a second element of the system, a quite definite third element of the same system is derived.

If a is the first element, b the second, and c the third, one writes symbolically

$$ab = c, \qquad c = ab,$$

and says that c is composed from a and b, and that a and b are the components of c.

In this composition the commutative law is in general not assumed; that is, ab may be different from ba. On the other hand,

2. the associative law is assumed,

that is, if a, b, c are any three elements of P, then

$$(ab)c = a(bc).$$

From this it follows, by complete induction, that one always obtains the same result if, in any finite row of elements of P, $a, b, c, d \ldots$, one first composes two adjacent elements, then again two adjacent elements, and so on until the whole row is reduced to one element. This element is denoted by $abcd \ldots$. Thus, for example,

$$\begin{aligned} abcd &= (ab)cd = [(ab)c]d = (ab)(cd) \\ &= a(bc)d = [a(bc)]d = a[(bc)d] \\ &= ab(cd) = (ab)(cd) = a[b(cd)]. \end{aligned}$$

3. It is assumed that, if $ab = ab'$ or $ab = a'b$, then necessarily in the first case $b = b'$, and in the second case $a = a'$.

If P comprises a finite number of elements, then the group is called finite and the number of its elements its degree.

For finite groups 1., 2., 3. imply the following consequence:

4. If two of the three elements a, b, c of P are given arbitrarily, then the third can always be determined, and in only one way, so that $ab = c$.

Indeed, if a, b are the given elements, the assertion 4. coincides with 1. If, however, a and c are given, let the second component b in the composite ab run through the whole system P, whose degree is n. Then by 1. and 3. the elements ab so obtained are all distinct elements of P, and since their number is n, all elements of P, and hence also c, must occur among them. The same argument applies if b and c are given, by letting a run through the whole system P.

For infinite groups this conclusion is no longer valid. Thus for infinite groups property 4. is also included as a requirement in the definition.

In the permutation groups considered in the first volume the marks of this general concept of a group are easily recognized.

We now draw from this definition some quite general consequences. By 4., for every given b there is an element e in P which satisfies

$$eb = b, \tag{1}$$

and this e is independent of b; for from (1) it follows, for every c, that

$$ebc = bc,$$

and by 4. the element bc may mean any element of P. Similarly there is an element e' satisfying, for every b,

$$be' = b. \tag{2}$$

But this element e' is not distinct from e; for, putting $b = e'$ in (1) and $b = e$ in (2), one obtains

$$ee' = e', \qquad ee' = e,$$

and hence $e = e'$.

The element e changes nothing when it is composed with arbitrary elements of P and is called the identity of the group. In many cases it may, without misunderstanding, simply be denoted by "1".

For each element a there is by 4. a definite element a^{-1} satisfying

$$a^{-1}a = e. \tag{3}$$

From (3), (1), and (2) it follows that

$$a^{-1}aa^{-1} = ea^{-1} = a^{-1} = a^{-1}e,$$

and consequently, by 3.,

$$aa^{-1} = e. \tag{4}$$

The two elements a, a^{-1} are called opposite or reciprocal to one another.

In special cases the commutative law may also hold for the composition of the elements of a group P; that is, for every two elements a, b of the group one may have $ab = ba$.

5. Groups which have this property are called commutative groups, or also Abelian groups.

If the elements of two groups

$$a, b, c, d \ldots \qquad \text{and} \qquad a', b', c', d' \ldots$$

correspond to one another in a one-to-one manner in such a way that whenever $ab = c$ also $a'b' = c'$, then the groups are called isomorphic. The evident theorem holds that two groups isomorphic with a third group are isomorphic with one another. One may therefore collect all mutually isomorphic groups into one class of groups, which is itself again a group, whose elements are the generic concepts obtained by collecting the corresponding elements of the individual isomorphic groups into a common concept. The individual mutually isomorphic groups are then to be regarded as different representatives of one generic concept.

The properties of groups which may come into consideration are of different kinds. They may attach to particular groups and be derived from the nature of the elements of which the group consists, or from the nature of the law of composition. Or they may attach to groups as such and then have to be derived solely from the definition of the group concept. The latter properties are common to all isomorphic groups and may be called invariant properties of the group. If a comparison is allowed, one may recall the difference between metric and projective properties in geometry.

To the first kind of properties, derived from the special nature of the elements, belong for example, in permutation groups, the properties of transitivity and intransitivity, of primitivity and imprimitivity. To the invariant properties belong commutativity or noncommutativity, the degree, the divisors and their index, and the normal divisors. We shall first deal with these invariant properties, for which it is immaterial from which representative of a class of isomorphic groups they are derived.

§ 2. The Divisors of Finite Groups.

As we have already seen in the first volume, it is a particularly important question whether a part of the elements of a group is itself again a group. For this it is necessary and sufficient that every two elements of this part, when composed, again give an element belonging to the same part. This smaller group is then called a divisor of the whole group. The identity element "1" by itself forms a group in every group, and is therefore a divisor in every group.

We now restrict ourselves, as almost always in what follows, to finite groups, and suppose that

$$P = 1, a_1, a_2, \dots, a_{n-1}$$

is a group of degree n, with elements $a_0 = 1, a_1, a_2, \dots, a_{n-1}$, and that

$$Q = 1, a_1, a_2, \dots, a_{v-1}$$

is a divisor of P of degree v. Every such divisor Q must have the fundamental properties of a group and therefore certainly contains the identity element "1", and with each of its elements a_i also the opposite element a_i^{-1}. If $v < n$, we call Q a proper divisor of P. Thus if b is an element of P not contained in Q, then the elements

$$Q_1 = b, a_1 b, a_2 b, \dots, a_{v-1} b$$

are all distinct from one another and are all not contained in Q; for if, say, $a_i b$ were contained in Q, then, since Q is a group, $a_i^{-1} a_i b = b$ would also have to be contained in Q, contrary to the assumption.

If the whole group P is not yet exhausted by Q and Q_1, we take one of the remaining elements, which we may denote by c, and form

$$Q_2 = c, a_1 c, a_2 c, \dots, a_{v-1} c.$$

Then the elements of Q_2 are all distinct not only from one another, but also from the elements of Q and Q_1. For if, for example, $a_i c = a_j b$, it would follow that $c = a_i^{-1} a_j b$; but $a_i^{-1} a_j$ is contained in Q and is therefore identical with one of the elements $1, a_1, a_2, \dots, a_{v-1}$, so that, contrary to the assumption, c would be contained in Q_1. We continue in this way, forming the systems $Q, Q_1, Q_2, \dots, Q_{\mu-1}$, the last of which is

$$Q_{\mu-1} = g, a_1 g, a_2 g, \dots, a_{v-1} g.$$

Then P must be exhausted by the systems $Q, Q_1, Q_2, \dots, Q_{\mu-1}$, and since these systems contain only distinct elements, it follows that

$$n = v\mu,$$

or the theorem:

1. The degree of a group is divisible by the degree of each of its divisors. The quotient $\mu = n : v$ is called the index of the divisor Q.

The systems $Q_1, Q_2, \dots, Q_{\mu-1}$ we call, as in the special case of permutation groups, the side groups belonging to Q, and denote them by

$$Q_1 = Qb, \quad Q_2 = Qc, \quad \dots \quad Q_{\mu-1} = Qg,$$

so that we may also write symbolically

$$P = Q + Qb + Qc + \cdots + Qg. \tag{1}$$

At times we shall also call the whole system $Q, Q_1, \dots, Q_{\mu-1}$ a system of side groups, and thus call the representation (1) the decomposition of P into a system of side groups.

From the manner in which the side groups are formed it follows further that, if b, c are any two elements of P, the two systems Qb and Qc are either wholly identical or have no element in common. For if $a_1 b = a_2 c$, where a_1, a_2 are two elements of Q, then for every other element a of Q one also has

$$aa_1^{-1}b = aa_2^{-1}c,$$

and when a runs through the whole group Q, each of aa_1^{-1} and aa_2^{-1} also runs through the whole group Q. Hence Qb and Qc are identical.

As in Vol. I, § 154, 2., we may also decompose P into side groups in the following way:

$$P = Q + bQ + cQ + \cdots + gQ.$$

The side groups do not have the marks of a group. For two elements of Q_1 to give, on composition, again an element of Q_1, one would have to have, say,

$$a_1 b a_2 b = a_3 b,$$

and hence $b = a_2^{-1} a_1^{-1} a_3$. Thus, contrary to the assumption, b would be contained in Q. The name side group is therefore to be understood only improperly.

Two divisors Q, Q' of P always have the element 1 in common. They may, however, also have other elements in common, and these common elements form a group, which we shall denote by R. For if the elements a and b belong both to Q and to Q', then, since both Q and Q' are groups, the same is true of the composite ab; hence R is a group. This common group we call the greatest common divisor of Q and Q', or, with a geometrical resonance, the intersection of Q and Q'. In the same way it follows that the elements common to any number of divisors of P form a group, which we likewise call the greatest common divisor or the intersection of all these groups.

In every group we can form divisors by the following procedure. The identity element by itself is always a divisor.

Denote the repeated composition of an element with itself by powers, and denote the identity element by a^0. Then the row of elements

$$1, a, a^2, a^3, \dots$$

all of which belong to P, cannot consist entirely of distinct elements. If the first element which occurs for the second time is

$$a^n = a^{n+\alpha},$$

then it follows that $a^\alpha = 1$, and that α is the least positive number satisfying this condition. Then, if $m = q\alpha$ is any multiple of α, one has $a^m = 1$; conversely, whenever $a^m = 1$, m must be divisible by α. For otherwise one could write $m = q'\alpha + \alpha'$, with α' positive and less than α, and then $a^{\alpha'} = 1$, contrary to the assumption that α is the least positive exponent for which $a^\alpha = 1$.

One has

$$a^{\mu} = a^{\mu'}$$

always and only when $\mu \equiv \mu' \pmod{\alpha}$; here the exponents may also be negative, if $a^{-\nu}$ denotes the ν-th power of the element a^{-1}, or the element opposite to a^{ν}.

The row

$$A = 1, a, a^2, \dots, a^{\alpha-1}$$

then contains α mutually distinct elements of P, in which all powers of a are represented. The elements A plainly form a group of degree α, since the exponents simply add under composition. This group is a divisor of P; hence α is a divisor of n. Thus for every element $a^n = 1$.

The group A is called the period of the element a, and α is also called the degree of the element a.

§ 3. Normal Divisors of a Group.

Let P be a group as above, let Q be a divisor of P, and let b be an element of P not contained in Q. Then the side group Qb is not a group. On the other hand the system $b^{-1}Qb$, or written out,

$$1,\ b^{-1}a_1 b,\ b^{-1}a_2 b, \dots,\ b^{-1}a_{v-1}b$$

certainly forms a group, since

$$b^{-1}a_1 b \cdot b^{-1}a_2 b = b^{-1}a_1 a_2 b,$$

and this group is isomorphic with Q. The group $b^{-1}Qb$ is called the group transformed by b. If b itself belongs to Q, then $b^{-1}Qb$ is identical with Q. If b is taken to be the different elements of P, one obtains a whole family of such groups, which we call the divisors of P conjugate to Q, or briefly conjugate groups.

It may happen that all conjugate divisors are identical with one another. We have already seen, in the case of permutation groups, that this case is of special importance, and therefore introduce the following general definition.

If Q is a divisor of P which is identical with all its conjugate divisors, then Q is called a normal divisor of P.

The group formed from the single element 1 is a normal divisor of every group. We obtain another normal divisor in the greatest common divisor R of all divisors conjugate, in P, to any divisor Q.

Indeed, it was proved above that R, as the intersection of several divisors of P, is a group. If

$$Q, Q', Q'', \dots \tag{1}$$

are the divisors of P conjugate to Q, and if b is any element of P, then the system of groups

$$b^{-1}Qb, \quad b^{-1}Q'b, \quad b^{-1}Q''b, \dots \tag{2}$$

is not different from the system (1). If R is the intersection of the groups (1), then $b^{-1}Rb$ is the intersection of (2), and consequently R is identical with $b^{-1}Rb$; that is, R is a normal divisor of P.

If N is a normal divisor of any group P and b is an arbitrary element of P, then

$$b^{-1}Nb = N \quad \text{or} \quad Nb = bN. \tag{3}$$

If P has degree n, if N has degree v and index μ, so that $n = \mu v$, then the elements $1, b_1, b_2, \dots, b_{\mu-1}$ may be chosen so that the decomposition of P into side groups gives

$$P = N + Nb_1 + Nb_2 + \cdots + Nb_{\mu-1} = N + b_1 N + b_2 N + \cdots + b_{\mu-1}N.$$

Thus

$$N_0 = N, \quad N_1 = Nb_1 = b_1 N, \quad N_2 = Nb_2 = b_2 N, \ldots, \quad N_{\mu-1} = Nb_{\mu-1} = b_{\mu-1} N \tag{4}$$

is the system of side groups.

If a is any element in N, then Na is identical with N; hence Nab is also identical with Nb. Since every element c of P occurs in N or in one of its side groups, and so is contained in the form ab_k, it follows that $Nc = cN$ must coincide with one of the systems (4).

We also mention the following theorem, whose correctness follows immediately from the isomorphism of transformed groups. If Q is a divisor of P and R is a divisor of Q, then, if Q' and R' are obtained from Q and R by transformation with the same element of P, R' is also a divisor of Q'; and if R is a normal divisor of Q, then R' is also a normal divisor of Q'.

§ 4. Composition of Parts.

If A and B are any two rows of elements of a group P–whether groups or not–we shall understand by the symbolic product AB the totality of all elements obtained by composing an element a of A with an element b of B, according to the rule prevailing in P, to form ab. We call this kind of composition of A and B into AB the composition of parts of P. We distinguish here between a part and a divisor of P, so that a divisor is always to be a group, whereas this is not necessary for a part.

In the same way one may compose three or more parts $A, B, C, \ldots$ of P, and the associative law holds for the composition of parts, as follows immediately from the fact that this law holds in P. Thus the meaning of a composite $ABC \ldots$ is uniquely determined.

In the composition AB one of the parts A, B may also consist of a single element, and then the notation AB agrees with the notation used above for side groups.

For the composition of parts the following theorems hold:

1. The necessary and sufficient condition that a part A of P be a group is the equation

$$AA = A. \tag{1}$$

For this equation says nothing other than that, if a and a' are contained in A, then aa' also belongs to A, and hence that A is a group.

2. The two equations

$$AA = A, \tag{2}$$

$$AB = BA, \tag{3}$$

where A is fixed and B may be any arbitrary part of P, express that A is a normal divisor of P.

Indeed, by (2) A is a divisor of P, and by (3), for each element b of P, one has $b^{-1}Ab = A$; that is, A is a normal divisor.

3. If A and B are two divisors of P (groups), then either of the two equations

$$AB = A, \qquad BA = A, \tag{4}$$

is the necessary and sufficient condition that B be a divisor of A.

For if B is a group and a divisor of the group A, and if a is contained in A and b in B, hence also in A, then ab is also contained in A. Thus A contains every element of AB. Since secondly B, as a group, contains also the element 1, AB contains every element of A; hence AB is identical with A. In the same way one sees that BA is identical with A. Conversely, from either of the equations (4), since A as a group contains the identity element, it follows that every element of B is contained in A, and hence that B is a divisor of A.

4. Under the composition of parts, the system of side groups belonging to a normal divisor of P itself forms a group, in which the normal divisor N is the identity, and Nb_k, Nb_k^{-1} are opposite elements.

For let N be a normal divisor of P and let

$$N, \quad N_1 = Nb_1, \quad N_2 = Nb_2, \dots \tag{5}$$

be the system of side groups. By the property of normal divisors N is commutable with every b_k, hence $b_h b_k N = N b_h b_k$, and consequently

$$N_h N_k = N N b_h b_k.$$

Since $b_h b_k$ is contained in P, $N b_h b_k$ is identical with one of the side groups (5); and since $NN = N$ may be put, it follows, if we put $N b_h b_k = N_i$, that

$$N_h N_k = N_i. \tag{6}$$

This proves the property § 1, 1. of groups for the composition of side groups. We have already emphasized that § 1, 2., the associative law, also holds. It remains to prove § 1, 3.; that is, if

$$N_h N_i = N_k N_l \tag{7}$$

is put, then from $N_h = N_k$ it follows that also $N_i = N_l$, and from $N_i = N_l$ it follows that also $N_h = N_k$.

According to the representation (5), however, we may write (7) in the two ways

$$N b_h b_i = N b_k b_l, \qquad b_h b_i N = b_k b_l N. \tag{8}$$

If now $N_i = N_l$, then we may also assume $b_i = b_l$, and from (8), by composition with b_i^{-1}, we obtain

$$N b_h = N b_k.$$

If $N b_h = N b_k$, then we assume $b_h = b_k$, and obtain from (8), by composition with b_h^{-1},

$$b_i N = b_l N,$$

and hence also $N_i = N_l$.

It is thereby proved that the system of side groups becomes a group under the composition of parts. That in this group the normal divisor N is the identity element follows immediately from 1., since then $NN_k = NNb_k = N_k$. And since $Nb_k N b_k^{-1} = N b_k b_k^{-1} = N$, the elements Nb_k and Nb_k^{-1} are opposite elements.

The group of the quantities N_k, whose existence has thereby been proved, is called the group of side groups, or the group complementary to N with respect to P. Following Holder, we denote it by P/N. The degree of the complementary group is equal to the index of the group N.

At the end of these considerations we mention one more theorem on normal divisors.

5. If N is a divisor of P, N' a divisor of N, and at the same time a normal divisor of P, then N' is also a normal divisor of N.

This is self-evident, since for every element b of P one has $b^{-1}N'b = N'$, and since every element of N is also an element of P.

But one must not conversely conclude that, if N is a normal divisor of P and N' is a normal divisor of N, then N' must also be a normal divisor of P.